Certificate

Name: HARSH KUMAR SAW Class:

Roll No: Exam No:

Institution _____

This is certified to be the bonafide work of the student in the _____ Laboratory during the academic year 20 / 20 .

No. of practicals certified _____ out of _____ in the subject of _____

.....
Teacher In-charge

.....
Examiner's Signature

.....
Principal

Date: ..... *Institution Rubber Stamp*

(N.B: The candidate is expected to retain his/her journal till he/she passes in the subject.)

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The Periodic Table

Remembering the order of the first 10 elements of the periodic table is child's play!

Just memorize the sentence given below. The chemical symbol of each of these elements is given in bold type.

Happy **He**nry **Li**kes **Be**ans, **B**rownies and **C**hocolate **Nu**ts **O**ver **F**riday's **Ne**ws

1																		18
1 H hydrogen	2											13	14	15	16	17		2 He helium
3 Li lithium	4 Be beryllium											5 B boron	6 C carbon	7 N nitrogen	8 O oxygen	9 F fluorine		10 Ne neon
11 Na sodium	12 Mg magnesium	3	4	5	6	7	8	9	10	11	12	13 Al aluminium	14 Si silicon	15 P phosphorus	16 S sulphur	17 Cl chlorine		18 Ar argon
19 K potassium	20 Ca calcium	21 Sc scandium	22 Ti titanium	23 V vanadium	24 Cr chromium	25 Mn manganese	26 Fe iron	27 Co cobalt	28 Ni nickel	29 Cu copper	30 Zn zinc	31 Ga gallium	32 Ge germanium	33 As arsenic	34 Se selenium	35 Br bromine		36 Kr krypton
37 Rb rubidium	38 Sr strontium	39 Y yttrium	40 Zr zirconium	41 Nb niobium	42 Mo molybdenum	43 Tc technetium	44 Ru ruthenium	45 Rh rhodium	46 Pd palladium	47 Ag silver	48 Cd cadmium	49 In indium	50 Sn tin	51 Sb antimony	52 Te tellurium	53 I iodine		54 Xe xenon
55 Cs caesium	56 Ba barium	57-71 lanthanoids	72 Hf hafnium	73 Ta tantalum	74 W tungsten	75 Re rhenium	76 Os osmium	77 Ir iridium	78 Pt platinum	79 Au gold	80 Hg mercury	81 Tl thallium	82 Pb lead	83 Bi bismuth	84 Po polonium	85 At astatine		86 Rn radon
87 Fr francium	88 Ra radium	89-103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadium	111 Rg roentgenium	112 Cn copernicium							

Key: atomic number / Symbol / name

57 La lanthanum	58 Ce cerium	59 Pr praseodymium	60 Nd neodymium	61 Pm promethium	62 Sm samarium	63 Eu europium	64 Gd gadolinium	65 Tb terbium	66 Dy dysprosium	67 Ho holmium	68 Er erbium	69 Tm thulium	70 Yb ytterbium	71 Lu lutetium
89 Ac actinium	90 Th thorium	91 Pa protactinium	92 U uranium	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

COMMON NAMES OF DIFFERENT CHEMICALS

Name	Common Chemical Name	Formula
Blue Vitriol	Copper sulphate	$CuSO_4.5H_2O$
Green Vitriol	Ferrous sulphate	$FeSO_4.7H_2O$
Epsom	Magnesium sulphate	$MgSO_4.7H_2O$
Plaster of Paris	Calcium sulphate hemihydrate	$CaSO_4.\ 1/2\ H_2O$
Gypsum	Calcium sulphate dihydrate	$CaSO_4.2H_2O$
Washing soda	Sodium carbonate	Na_2CO_3
Common salt or Rock salt	Sodium chloride	NaCl
Nitre or Salt Petre	Potassium nitrate	KNO_3
Baking soda	Sodium bicarbonate	$NaHCO_3$
Chalk	Calcium carbonate	$CaCO_3$
Caustic lunar	Silver nitrate	$AgNO_3$
Quick lime	Calcium oxide	CaO
Slaked lime or Milk of lime	Calcium hydroxide	$Ca(OH)_2$
Potash Alum	Potassium aluminium sulphate dodecahydrate	$K_2SO_4Al_2(SO_4)_3.12H_2O$
Bleaching powder	Calcium oxychloride	$CaOCl_2$
Carbolic Acid	Phenol	C_6H_5OH
Talc	Magnesium silicate	$MgSiO_3$
Water glass	Sodium silicate	Na_2SiO_3

CONVERSION FACTORS TO SI UNITS

Physical quantity	CGS	SI
Force	1 dyne	$= 10^{-5}$N(newton)
Pressure	1 Ba	= 0.1 Pa (Pascal)
Energy	1 erg	$= 10^{-7}$J(Joule)
Heat	1 cal	= 4.148J
Volume	1 Litre	= 0.001 m^3

STRENGTH OF AQUEOUS SOLUTION OF SOME CONCENTRATED ACIDS AND BASES

Compound	Molecular formula	Molecular mass	Approximate			Volume required to make 1ℓ of approximate 1M solution (ml)
			% of solute by weight	Specific gravity	Molarity	
Hydrochloric acid	HCl	36.5	36	1.18	11.65	85.8
Nitric acid	HNO_3	63.0	70	1.42	15.8	63.3
Sulphuric acid	H_2SO_4	98.0	98	1.84	18.4	54.3
Acetic acid	CH_3COOH	60.0	99.5	1.05	17.5	57.5.
Phosphoric acid	H_3PO_4	98.0	85.5	1.71	14.8	67.5
Sodium hydroxide	NaOH	40.0	50.5	1.53	19.4	51.5
Ammonia	NH_3	17.0	25.1	0.91	13.4	74.6

WEIGHT OF SUBSTANCES REQUIRED TO PREPARE ONE LITRE OF A SOLUTION OF A PARTICULAR NORMALITY

Sr. No.	Substance	Mol. Mass	Eq. Mass	Quantity in grams for solution of strength			
				N/10	N/20	N/30	N/40
1.	Oxalic acid, $H_2C_2O_4.2H_2O$	126.08	63.04	6.3040	3.1520	2.1013	1.5760
2.	Hydrochloric acid, HCl	36.50	36.50	3.650	1.8250	1.2167	0.9125
3.	Sulphuric acid, H_2SO_4	98.00	49.00	4.9000	2.4500	1.6333	1.2250
4.	Acetic acid, CH_3COOH	60.00	60.00	6.0000	3.0000	2.0000	1.5000
5.	Sodium carbonate, Na_2CO_3	106.00	53.00	5.3000	2.6500	1.7667	1.3250
6.	Sodium hydroxide, NaOH	40.00	40.00	4.0000	2.0000	1.3333	1.0000
7.	Sodium bicarbonate, $NaHCO_3$	84.00	84.00	8.4000	4.2000	2.8000	2.1000
8.	Calcium carbonate, $CaCO_3$	100.00	50.00	5.0000	2.5000	1.6667	1.2500
9.	Potassium dichromate, $K_2Cr_2O_7$	294.20	49.03	4.9030	2.4515	1.6343	1.2257
10.	Potassium permanganate, $KMnO_4$	158.03	31.60	3.1600	1.5800	1.0533	0.7900
11.	Mohr's salt $FeSO_4(NH_4)_2SO_4.6H_20$	392.10	392.10	39.2100	19.6050	13.0700	9.8025
12.	Sodium thiosulphate, $Na_2S_2O_3.5H_2O$	248.20	248.20	24.8200	12.4100	8.2733	6.2050

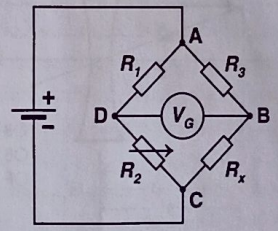
Wheatstone bridge

In the figure, $\mathbf{R_x}$ is the unknown resistance to be measured; $\mathbf{R_1}$, $\mathbf{R_2}$ and $\mathbf{R_3}$ are resistors of known resistance and the resistance of R_2 is adjustable. If the ratio of the two resistances in the known leg (R_2/R_1) is equal to the ratio of the two in the unknown leg (R_x/R_3) then the voltage between the two midpoints (**B** and **D**) will be zero and no current will flow through the galvanometer $\mathbf{V_g}$. If the bridge is unbalanced, the direction of the current indicates whether R_2 is too high or too low. R_2 is varied until there is no current through the galvanometer, which then reads zero.

At the point of balance, the ratio of

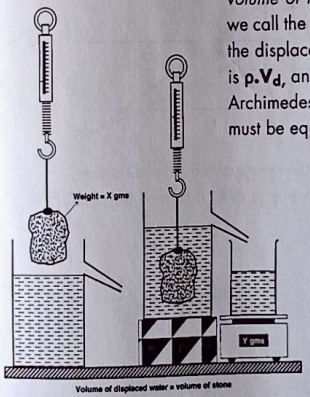
$$\frac{R_2}{R_1} = \frac{R_x}{R_3}$$

$$\Rightarrow R_x = \frac{R_2}{R_1} R_3$$



Archimedes Principle

To determine the magnitude of the buoyant force that a liquid exerts on a submerged object, we must consider how much of the object (of total volume V) lies below the liquid surface. Since only the submerged portion of the object displaces liquid, we see that *the displaced volume of liquid is equal to the object's submerged volume.* If we call the difference in density of the fluid and object as ρ and the displaced volume $\mathbf{V_d}$, then the mass of the displaced liquid is $\mathbf{\rho.V_d}$, and the *weight* of the displaced liquid is $\mathbf{\rho.g.V_d}$. From Archimedes' principle, the magnitude of the buoyant force B must be equal to this weight:

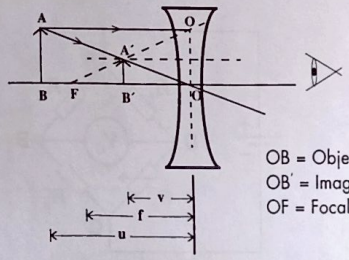


$$\mathbf{B=\rho.g.V_d}$$

For an object *completely* submerged in the liquid, V_d=V. However, for an object only *partially* submerged, V_d<V.

Focal length for concave lens

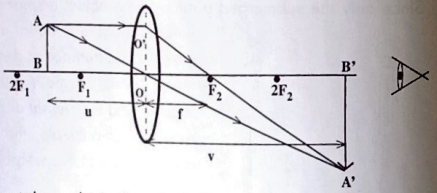
Consider an object **A B** placed at a distance of **u** in front of a concave lens of focal length " **f** " on the principal axis of the lens. Concave lens forms a virtual and erect image **A'B'** at a distance of " **v** " from the optical centre of the lens as shown in the diagram below.



OB = Object distance = u
OB' = Image distance = v
OF = Focal length = f

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Focal length of convex lens



Let **AB** represent an object placed at right angles to the principal axis at a distance greater than the focal length **f** of the convex lens. The image **A'B'** is formed beyond $2F_2$ and is real and inverted.

OB = Object distance = u
OB' = Image distance = v
OF_2 = Focal length = f

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Date _____

Expt. No. 01 Page No. 01

Introduction, Bio medical Waste Management

Clinical biochemistry deals with the biochemical aspects of diseases.

Various biochemical tests done in the laboratory helps the clinical in diagnosis, treatment and monitoring of the disease process. The qualitative & Quantitative tests also help in screening & preventing the diseases as well as in forensic science.

The accuracy & reliability of any biochemical test determine its utility to the clinician. This can be achived easily if the laboratory personal are well versed with the operation of equipments preparations of reagent & the methodologies of various tests. a working knowledge of these is mandatory in a biochemistry laboratory.

Hazard in clinical Biochemistry Laboratory:

The advances in clinical practice has led to a steady increase in the no. and range of investigations performed in the clinical biochemistry laboratory and consequently to an increase in various hazards.

Hazards arise from 3 main basic sources:

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1) From dangerous chemicals

2) From inspected specimens sent for analysis &

3) From faulty apparatus & instruments.

These are further increased by carelessness, untidiness, faulty hygiene, conduct of staffs & unsatisfactory buildings and working conditions. The hazards could be physical, chemical or infective in nature.

Physical:

Electrical shocks, radiation exposure, physical trauma, from broken glass ware, needle etc.

Chemical:

Direct contact with skin, lips, mouth. Aspiration of corrosive toxic chemicals leading to damage to stomach, liver, skin etc. Inhalation of toxic vapour leading to lung damage. Explosives and inflammable chemicals can lead to burns or fire.

Infection:

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Specimen used in the laboratory are all potentially infected. The main risk beneficiary of viral hepatitis & AIDS Cardinal safety rules and precautions:

Overalls and gloves must be worn at a routine. Hair must be tied up. One should not eat drink or smoke in working area. Strong acids/alkalies or toxic chemicals should not be mouth pipette, instead used dispensers, bulbs or automatic pipettes. Hand must be washed with soap after work and before meals. The used gloves tips unused samples, glassware must be properly discarded after use.

Good Housekeeping:

Proper storage and labling of chemicals must be done. Work area must be clean and free of unwanted reagents/glassware to avoid littering. Fire extinguishers must be in a good, ready and working condition. Sterilization and disinfection must be done to avoid contamination and infection.

This can be done by:

1) Boiling any instrument in 10 min.

2) Soaking for 20 min in disinfectant solution

Sodium hypochlorite 1%

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Glutaraldehyde 2%
Lysol 2.5 %
Savlon 5%

- All the equipment must be earthed for electrical safety and proper instructions should be followed before use.
- First aid box should be trained to use it.
- All needles, syringe must be destroyed to prevent their reuse.

First Aid in the Laboratory:

1) Skin burns should be washed under running water or iced water and petroleum jelly or any burn ointment should be applied and then covered with sterile gauze.

2) Chemical injury to eyes must be treated by thorough washing with water and then putting 2% sodium carbonate drops till referred to a specialist.

3) Inhalation injury by toxic fumes is best treated by removing the person to an uncontaminated atmosphere. Irritation to throat can be smoothened with hot water vapour inhalation or a warm drink.

4) In accidental swallowing of chemicals the mouth must be thoroughly rinsed with water. If complete swallowing has occured the

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Bio-medical Disposal Guidelines

Cat.	Type of Bag/Container Used	Types of Waste
Yellow 	Non-chlorinated plastic bags Separate collection system leading to effluent treatment system	(a) Human Anatomical Waste (b) Animal Anatomical Waste (c) Soiled waste, (d) Expired or Discarded Medicines (e) Chemical Waste (f) Micro, Bio-t and other clinical lab waste (g) Chemical Liquid Waste
	Treatment/Disposal options	
	Incineration or Plasma Pyrolysis or deep burial	
Red 	Non-chlorinated plastic bags or containers	Contaminated Waste (Recyclable tubing, bottles, intravenous tubes and sets, catheters, urine bags, syringes (without needles) and gloves
	Treatment/Disposal options	
	Auto/Micro system and then sent for recycling or sent to landfill	

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person should be made to drink water followed by milk in case of acids and lemon juice/vinegar in case of alkalis.

5) Injuries caused by broken glass are best treated by washing the wound immediately to remove any glass piece, applying monosaccharide/acriflavine ointment & covering with gauze & tape. Heavy bleeding must be stopped by compression or tight bandage before sending the patient to the emergency.

6) Contamination with infected material, in wounds caused by broken glass wire, must be thoroughly rinsed with water and washed with soap solution before antiseptic solution.

Disposal of laboratory waste:

⇒ Bio medical waste is the waste which is generated during the diagnosis, treatment or immunization of human beings or animals or in research activities.

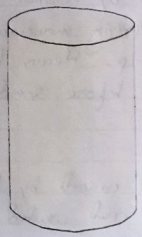
⇒ To avoid the risks of biological hazard safe handling and disposal of these infect materials is required.

⇒ Waste generated in the lab could be enormous in quantity & variety for ex: left over samples, needles, syringes, obsolete articles of various nature etc.

⇒ There are guidelines to dispose them after taking due care.

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Bio-medical Disposal Guidelines

Cat.	Types of Bag/ Containes used	Types of Waste	Treatment/ Disposal option
White 	(Translucent) Puncture, Leak, temper proof containes	Waste sharps including Metals	Auto clave/ Dry Heat sterilization followed by shredding or Mutilation or encapsulation
Blue 	Cardboard boxes with blue colored marking	Glassware	Disinfection or auto/Microwave/ hydro and then sent for recycling
	⇒ Black Bag (Expired or discarded medicine)		

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⇒ It is recomended that waste should be segregated at the point of generation & disposed in bags with colour coding

Black Bag for Municipal Dumps:

⇒ it includes non infective biodegradable waste like papers, peels, wrappers etc.

Yellow Bag for Incinerator:

It includes cotton swabs, left over samples, gloves, items contaminated with blood & other samples, discarded medicines

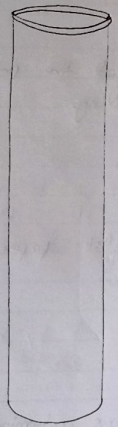
Blue bag for shredder:

It includes syringes, Needles, etc. which are first destroyed in a needle destroyer. Other sharp objects are first put in 1% bleaching powder and then segregated in puncture proof cardboard boxes.

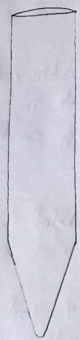
All bags and containers are properly closed and labeled with water proof ink or by self adhesive labels

Items soiled with blood must first be soaked in 1% sodium hypochlorite solution using a double cage basket and then washed

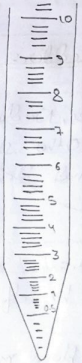
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Test tube



Non-graduated centrifugal tube



Graduated centrifugal tube

Date _____

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Glassware Introduction

Aim: Introduction about glassware

- The most useful glassware used in biochemistry lab is made up of borosil, because it has low alkaline earth content and free of contaminants like lead and mercury.

1. Test tube :-

- It is the U-shaped tube made up of thin glass without marking.
- It has non-uniform diameter throughout its length used for holding small quantities of substances, used for qualitative observations on heating.

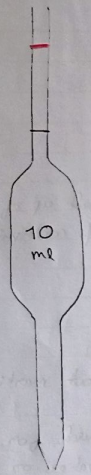
2. Cuvette :-

- It is similar to test-tube but small in size
- It has similar diameter throughout the length
- It has well-defined optical activity
- Mainly it is used in quantitative analysis for measuring optical activity.

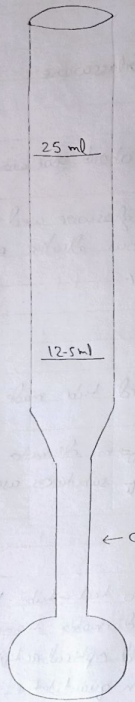
3. Centrifugal tube :-

- It is similar to that of a test-tube but the lower end is tapered.

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pipette



Folin wu tube

Date

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- It is of 2 types: 1) Graduated centrifugal tube (with markings)
  2) Non-graduated centrifugal tube (without markings)
- It is mainly used for separation of serum or plasma from blood

4. Pipette :-

- There are 3 types of pipettes generally used in biochemistry lab :-

(a) Simple pipette: Usually of 5-10 ml of capacity, the fluid is sucked and is transferred to another glassware. It is used for titration.

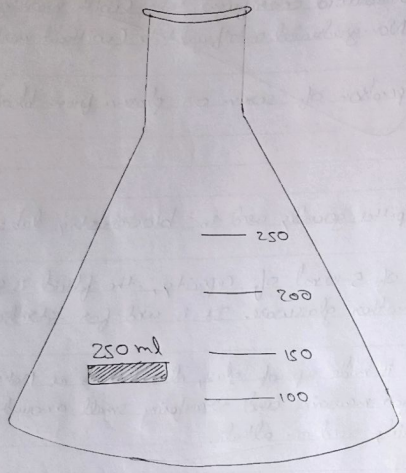
(b) Graduated pipette: It is made up of glass, also known as Mohr pipette. It is used for measuring and transfering small amount of biochemical reagent, mainly acid or alkali.

(c) Bulb pipette: Very sensitive to temperature changes, measurement is generally taken out at 273K. It comes in the capacity of 5, 10, 20 ml and is generally used in titration.

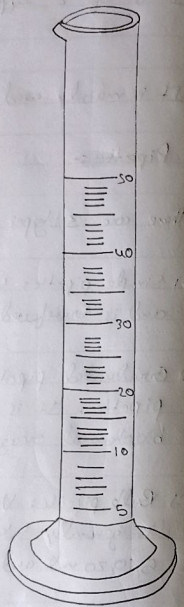
5. Folin wu tube :-

- It is a broad test tube having narrow constriction near the end and contain bulb at the base.
- It has 2 markings 12 ml and 25 ml.
- Changes in atmospheric pressure doesn't affect this tube because

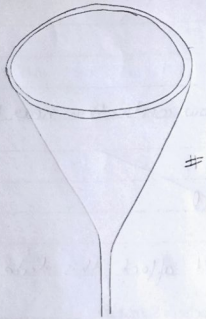
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Conical flask



Measuring cylinder



Funnel

Date

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of constriction, so it is simple than a simple tube used for estimation of sugar in a given sample

6. Conical Flask:-

- It has upper cylindar portion which ends on broad base.
- It is of different capacities like 100ml, 150ml, 200ml, 400ml, etc Mainly used for titration and applying biochemical tests

7. Measuring Cylinder:-

- It has cylindrical upper and broad base at lower end.
- It is graduated and has uniform markings, generally used for preparation of different types of biochemistry reagents

FUNNELS:-

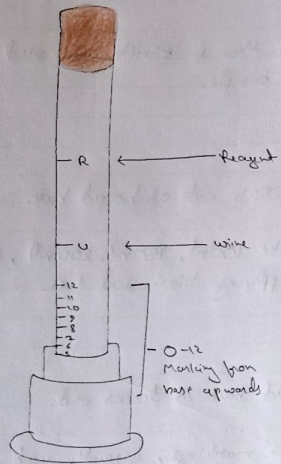
Mainly 3 types of funnels are used in biochemistry lab:-

(a) long stem funnel - It is used for transfer liquids into narrow necked bottles.

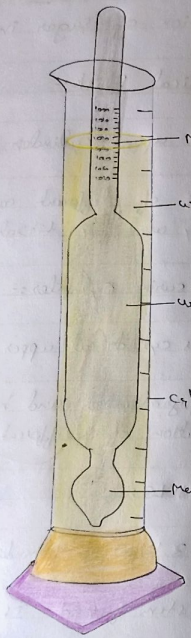
(b) Short stem funnel - It is used for transfer liquid from one container to another.

(c) Seperating funnel - It has a glass stopper. It is open at one end

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Esbach's Albuminometer



Urinometer

Date

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& stop cock at another end. It is used to separate two immiscible liquids or solvents

8. Esbach's Albuminometer :-

- It is specialised test tube with rubber cork at upper end and closed at lower end has separate marking upto 12
- It mainly used for quantitative estimation of albumin in a given urine sample.

9. Urinometer :-

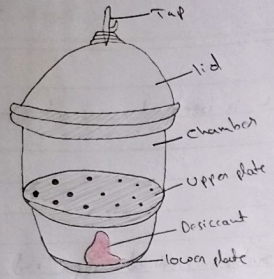
- It is used for measuring the specific gravity in urine sample.
- Upper grade marking is upto 1.000
- It has 2 parts, upper part is constricted, lower part is expanded with/without mercury bulb at base
- Urinometer is dipped in urine sample upto marking then it gives the reading of specific gravity of urine sample.

Note: care must be taken, urinometer must not touch the walls of container.

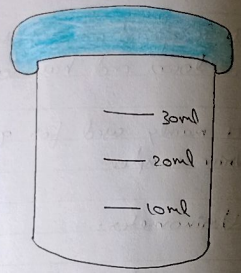
10. Desiccator

- Desiccator of seal enclosure.

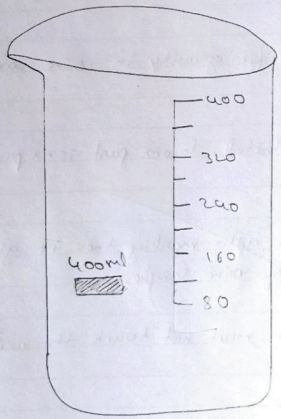
Teacher's Signature



Dessicator



Urine Jar



Beaker

Date

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or packed in yellow bags for incineration.

- It is used for preventing moisture sensible items ~~Inter~~
- It protects chemicals which are hygroscopic.

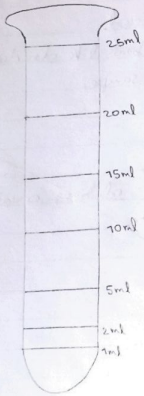
11. Urine Jar :-

- It is a wide mouth conical vessel.
- The upper part of the jar is beak like structure and lower has circular end
- It is used to store urine sample.

12. Beaker :-

- Used for mixing 2 or more solutions available in different capacities like 100ml, 150ml, 200ml etc.

Teacher's Signature



Calibrated test tube

Date

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Calibration of test tube

Apparatus Required: Test tube, pipettes, beakers, burettes, distilled water

Procedure:

(1) Take burette and fix it into the stand.

(2) Fill the burettes with distilled water upto the mark.

(3) Check the lower meniscus.

(4) Take test tube of 1ml of water from burette to test tube. Mark the level of 1ml with a diagram of the test tube.

(5) Repeat the procedure for 2ml, 5ml, 10ml and mark the levels

(6) Measure the capacity of the test tube using aforementioned steps.

Results: The capacity of the given test tube is 25 ml.

Precautions:

(1) Prevent bubbles from forming in the pipette.
(2) Markings should be done considering lower meniscus of water.
(3) Burette should be correctly positioned.

Teacher's Signature

26/10/26

Composition of molisch's test :- Alcoholic alpha napthol

Principle :- Carbohydrates undergo dehydration by H_2SO_4 by releasing 3 molecules of H_2O to form hydroxy methyl furfurals, which will couple with alcoholic alpha napthol present in the Molisch's reagent to form violet colour ring which is seen at the junction of two liquids.

Pentose sugars give furfural derivatives.

This test is to detect the carbohydrate is present in the solution

Composition of Benedict's Reagent :-

Cupric Sulphate : Provides copper ions
Sodium Carbonate : Gives alkaline nature
Sodium citrate : acts as chelating agent

Principle :- Carbohydrates undergo tautomerisation to form enediols in alkaline medium provided by the sodium carbonate. These enediols are the powerful reducing agents reduce cupric ions to cuprous ions. On heating it will convert to cuprous oxide give a red colour ppt.

This test is also called as semi quantitative test in abnormal urine and to detect the sugar present in the urine.

Basing upon the colour of the test result we can know how much % of sugar is present in the urine sample.

Blue - 0gm% or nil, (-)
Green - 0.1 - 0.5 gm% / (+)
Yellow - 0.5 - 1 gm% / (++)
Orange - 1 - 1.5 gm% / (+++)
Red - 1.5 - 2 gm% / (++++)

Date

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Carbohydrate solution identification

Aim: Identification of given carbohydrate solution.

Apparatus: Test tubes, chemical reagents, test tube stand, Bunsen burner, droppers, holder.

Physical Examination:

Appearance -
Colour -
Odour -
Solubility -
Reaction to litmus -

Chemical Examination:

S.No.	Test	Observation	Inference
1.	Molisch's test Take 2ml of given solution, add 5 drops of molisch reagent and add 2ml of conc. H_2SO_4 along the slides of the test tube.	Violet colour ring is observed at the Junction of 2 liquids	This test is specific for carbohydrate

Teacher's Signature

Composition of Barfoed's reagent :-

Glacial acetic acid - Provides acidic medium
Cupric acetate - Provides copper ions

Principle : Carbohydrate undergo tautomerisation to form enediols even in the acidic medium provided by the glacial acetic acid. These enediols are the powerful reducing agents reduce cupric ions to cuprous ions. On heating it will convert to cuprous oxide give a red colour ppt.

Composition of Seliwanoff's reagent :- Resorcinol and HCl.

Principle :- Carbohydrates undergo dehydration when react with conc. HCl by releasing 3 molecules of H_2O to form hydroxy methyl furfurals, which will condense with resorcinol present in the seliwanoff's reagent to form cherry red colour.

Principle of Iodine test :-

Test depends upon the property of adsorption, absorbing small iodine molecules to form blue coloured complex.

Note :- Blue colour disappears upon heating and reappears on cooling.

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S.No	Test	Observation	Inference
02.	Benedict's test Take 5 ml of benedicts reagent, add 8 drops of given solution, heat it for 2 min	Appearance of brick red colour ppt. seen in the test tube	Reducing sugar are present
03.	Barfoed's test Take 2 ml of solution, add 1 ml of barfoeds reagent, heat it for 2 min. Then cool the test tube under running tap water.	Red colour scum is adherent to the sides and bottom of the test tube	Monosaccharides are present
04.	Seliwanoff's test Take 3 ml of seliwanoff's reagent, add 1 ml of given solution. Heat it for 30 sec.	Cherry red colour is observed	keto sugar (Fructose) is present
05	Iodine test Take 2 ml of starch solution add few drops of iodine.	Deep blue colour is observed	Starch is present

Result : The given solution contains reducing monosaccharides is glucose

Teacher's Signature

Test for urea (Urease test) :-

⇒ In acidic medium urease reacts on urea form coloured substance.

Test for uric acid (Schiff's test) :-

⇒ Uric acid reduces ammonical $AgNO_3$ to metallic Ag which is black.

Test for creatinine (Jaffe's test) :-

⇒ In alkaline medium creatinine reacts with picrate and forms orange red coloured creatinine picrate.

Urine is an ultra filtrate formed by the kidneys containing waste and toxic substance but kidneys reabsorbs certain essential substances.
Eg. ⇒ Glucose, electrolytes, etc.

Constituents :-

01. Water
02. Inorganic constituents :- Na^+, k^+, Ca^{2+}, Mg^{2+}, Cl^-, $H_2PO_4^-$, HPO_4^{2-}, SO_4^{2-}
03. Organic constituents :- Urea, Uric acid, creatinine, Urobilinogen and Urochrome etc.

The most important pigment in wine are urochrome and urobilinogen. Urochrome is a compound of Urobilin and Urobilinogen. The colour of urine is due to Urochrome.

Complete urine analysis :-

Includes bacteriological analysis, Microscopic analysis, Biochemical analysis.

Date _____

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Constituents of Urine

Aim: To study the normal constituents of Urine.

Physical Examination :-

Volume - 60 ml provided (Avg. Volume - 1.5 L/day)
Appearance - Clear
Colour - Yellow colour
Odor - Aromatic
Ph - Slightly acidic
Sp. gravity - 1.010 - 1.024

Test for inorganic constituents :-

S.No	Experiment	Observation	Inference
01.	Test for ammonia :- Take 3 ml of urine sample, make it alkaline by adding 5% NaOH and boil it. Place a red litmus paper on the mouth of test tube.	Litmus paper turns blue	Indicates the presence of NH_3
02.	Test for carbonate :- Take 2ml of urine solution and add few drops of conc. HCl	Effervescence of CO_2	Indication of the presence of carbonates.

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Collection of Urine Sample :-

Mid stream morning sample, 24 hour urinary sample, Random urine sample.

Preservation of Urine Sample :-

Conc. HCl (10ml), Thymol (5ml of 100g/solution in isopropanol)

Physical properties :-

Volume ⇒ 60 ml provided [Avg. volume - 1.5L/day (range to 2.5L)]
- polyuria > 2.5 L/day
- oliguria < 300 ml/day
- Anuria - No urine output in 24 hours

Appearance :- Clear, transparent without any turbidity or sediment. However variation can occur and may point to many diseases

Colour :- Yellow colour or straw coloured
- Deep yellow - Jaundice
- Red - Haematuria

Odour :- Aromatic
- Fruity - ketoacidosis due to acetone
- Foul - Bacteria
- Mousy - Phenyl ketonuria

Ph :- Slightly acidic (4.7-8.0, Avg. 6.0), changes observed in physiological and pathological conditions of patient.

Sp. gravity :- 1.012 - 1.024

Date

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S.No	Experiment	Observation	Inference
03.	**Test For Chloride :-** Take 3ml of Urine sample and add 1ml of HNO_3 (conc) and 2% of $AgNO_3$	White ppt of AgCl	Presence of chloride
04.	**Test for Calcium :-** Take 3ml of urine sample and add 5 drops of 5% NaOH, add 2ml of saturated ammonium Oxalate solution and acidify with 1% HNO_3	White ppt is formed due to Calcium oxalate	Presence of calcium Indication
05.	**Test for Magnesium :-** Take 2-3ml of Urine sample and add 5 drops to 5% NaOH and 2-3 ml saturate ammonium Oxalate solution. Filter the above solution and add liquid NH_3. Mix well and allow to stand for few minutes	White ppt is formed	Indicate the presence of Magnesium
06.	**Test for Sulphur :-** Take 2-3ml of Urine and add 2ml of dil. HCl and 2ml of 10% $BaCl_2$ solution	White ppt is formed	Presence of Sulphur.

Test for Organic constituents :-

Teacher's Signature

⇒ It is determine by an instrument called urinometer. Urinometer consist of thin stem at the upper part and bulb at the bottom. Stem is graduated from 1000-1060 corresponding to specific gravities of 1.00 - 1.06 values which were calibrated at 15°C. The bulb at the bottom contains some fixed amount of mercury.

Take a clean round measuring cylinder. Fill it with urine to about 3/4th of its length. Gently place the urinometer into the cylinder and allow it to dip. Care should be taken that it does not touch the walls of cylinder. By virtue of the weight of the mercury, it dip into the urine to an extent dependant on the specific gravity of urine. The level of urine corresponding to the markings indicates the specific gravity. As it is calibrated at 15°c and therefore temperature correction is applied depending upon the actual temperature. A correction factor of one is added or subtracted for every 3°c rise or fall of temperature over the standard temperature of 15°c respectively.

Sp. gravity Calculation (Eg:-1)

Suppose observed sp gr. = 1.010
Suppose temp. of urine = 36°c
Temp. correction = 36°c - 15°c ⇒ 21°c
Factor of one is added or subtracted for every 3°c rise or fall ⇒ $^{21}/_{3}$ ⇒ 7

⇒ 1.010 + 0.007
Calculated Sp. gr. ⇒ 1.017

Sp. gravity Calculation (Eg-2):-

Suppose observed sp. gr. = 1.012
Suppose temp of urine = 9°c
Temp. correction = 9°c - 15°c ⇒ -6°c
Factor of one is added or subtracted for every 3°c rise or fall. ⇒ $^{-6}/_{3}$ ⇒ -2
Calculated sp. gr. ⇒ 1.012 - 0.002 ⇒ 1.010

Date

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S.No.	Experiment	Observation	Inference
01.	Test for urea :- (Urease test)		
	Take 2ml of urine sample and add 4 drops of phenolphthalein indicator till pink colour is observed make it slightly acidic with 1% acetic acid. So that colour disappears. Add a pinch of Urease powder keep it at room temperature	Pink colour is observed	Indicates the presence of Urea.
02.	Test for Uric acid :- (Schiff's test)		
	Take 2 ml of urine sample add 2ml of 2 ml sodium carbonate solution pour the mixture on the filter paper and moist it with $AgNO_3$ solution	Black stain is observed	Indicates the presence of Uric acid.
03.	Test for Creatinine :- (Jaffe's test)		
	Take 3ml of Urine sample and add 1% picric acid solution and 1 ml of 40% NaOH mix well and stand for a few minutes	Orange red colour is observed	Indicates the presence of Creatinine

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Inorganic Constituents :-

Ammonia :- Absence in urine sample. Urine on prolonged standing because of action of microorganisms urea is converted in ammonia.

Ammonia is not just a waste product of nitrogen metabolism. It is involved for the synthesis of many compounds like non essential amino acids, purine, pyrimidinines, amino sugars, etc. Ammonium ions are very important to maintain acid-base balance of body.

Ammonia toxicity is due to increased levels of blood ammonia conc. (Normal range 15-45 µg/dl) which is harmful to the brain results in slurring of speech and coma and finally death, if not corrected. To treat hyper ammonia intravenous administration of sodium benzoate and phenyllactate is done.

Cations like Na^+, K^+, Ca^{2+}, Mg^{2+} are excreated as salts in association with anions like SO_4^{2-}, Cl^-, PO_4^{3-}.

Phosphates :- The phosphates in the urine are mainly derived from the phospholipids. Two types:

1. **Alkaline phosphates :-** Phosphates of sodium, potassium and ammonium and they are derived from ingested inorganic phosphates.

2. **Earthy phosphates :-** Phosphates of calcium and magnesium and they are derived from ingested phospholipids and nucleoproteins (organic phosphates)

Urinary phosphates increase in acidosis and decrease in diarrhea.

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Sulphates :- Sulphate is ingested through sulfur containing amino acids and sulfur containing carbohydrates (chondroitin sulphate etc) Two types :-

1. **Inorganic sulfur :** (75-85%) Sulphur coith sodium, potassium ions.
2. **Ethereal sulfur:** (5-25%) Sulfur with organic compounds.

Organic constituents : NPN substances (Non protein Nitrogenous substances : Urea, Uric acid and creatinine).

Urea : End product of protein metabolism and increases with high protein diet.

Uric acid : End product of purine metabolism and increases in Gout and leukemia.

Creatinine : Catabolic waste product of tissue proteins. Creatinine is the anhydrate form of creatine

Creatinine coefficient : It is defined as the amount of creatinine excreted in 24 hrs urine sample / kg body weight.

Male : 20-26 mg/kg BW/24 hrs.
Female : 14-20 mg/kg BW/24 hrs.

Normal values of inorganic constituents in urine :

Ammonia ⇒ 0.7 g/day
Phosphorous ⇒ 1-1.5 g/day
Potassium ⇒ 2-2.5 g/day

Chloride ⇒ 10-15 g/day
Sulphur ⇒ 60-120 g/day

Calcium ⇒ 0.1-0.3 g/day
Sodium ⇒ 3-5 g/day

Normal values of organic constituents :-

Urea - 25-30 g/day
Uric acid - 0.5-0.8 g/day
Creatinine - 1.0-1.5 g/day.

Teacher's Signature

1. Benedict's test :-

Principle : Reducing sugar have free aldehyde or ketone group which reduces metallic ions, present in urine reduce alkaline cupric sulphate to coloured cuprous oxide upon boiling.

Date

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Abnormal constituents of Urine

Aim :- To detect the presence of abnormal constituents in the given urine sample.

Apparatus and Reagent Required :-

⇒ Test tubes, test tube holder, test tube stand
⇒ Spirit lamp / water bath
⇒ Droppers, pipettes, filter paper, funnel
⇒ Litmus paper / urinometer
⇒ Chemical reagents.

Physical Examination :-

1. Colour -
2. Appearance -
3. Odour -
4. Specific gravity -
5. Reaction with litmus (pH) -

Chemical Examination :-

1.	Benedict's test	Observation	Inference
	Take 5ml of benedict's reagent, add 7-8 drops of urine. Mix and boil vigorously for 2 min cool	Brick red / Yellow / Green precipitate formed.	Glucose is present

Teacher's Signature

2. Rothera's test :-

Principle: Ketone (acetone acetoacetic acid) react with sodium nitroprusside alkaline medium to form a purple coloured complex.

3. Heat coagulation test :-

Principle: Heat causes denaturation and precipitation of proteins (Albumin/globulin) in acidic medium.

4. Benzidine test :-

Principle: Heme acts as a peroxidase catalyst, decomposing H_2O_2 to release nascent oxygen which oxidizes benzidine, to form blue-green coloured product

5. Hay's Sulphur test :-

Principle: Bile salts lower the surface tension of urine, causing the sprinkled sulphur powder to sink instead of floating.

6. Fouchet's test :

Principle: Bilirubin is oxidized by fouchet's reagent (Ferric chloride + trichloroacetic acid) to green biliverdin.

Date

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2.	Rothera's test To 2-3ml urine, saturate with ammonium sulphate add sodium nitroprusside + shake + liquor ammonia from the side of the test tube.	Observation Ruby-red or purple ring form at the junction.	Inference ketone bodies are present
3.	Heat Coagulation test Fill 3/4 of test tube with sample + heat upper part, coagulation is formed. Add 2-3 drops of acetic acid.	Observation Coagulum is formed	Inference Protein is present
4.	Benzidine test Take 2 ml urine add 2-3 drops of benzidine reagent and a few drops of H_2O_2	Observation Immediate appearance of a deep blue green color	Inference Blood is present
5.	Hay's Sulphur test Take 5ml of urine in a clean test tube, sprinkle a pinch of powder of sulphur on top	Observation Sulphur powder sinks to the bottom	Inference Bile salts are present
6.	Fouchet's test Take 5ml urine add 1ml barium chloride filter, the precipitate on filter paper, add 1-2 drops of Fouchet's reagent	Observation Green or bluish green color develop on filter paper.	Inference Bile pigments are present

Result :-

The given urine sample has been analyzed and all the abnormal constituents are present

Teacher's Signature